VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

| Reg. Number: | 23BCE1463 |
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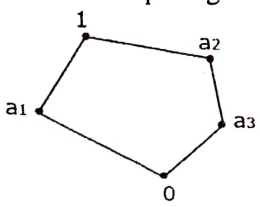
## Continuous Assessment Test (CAT) – II OCTOBER 2025

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|---|---|---|---|---|---|
| Programme | : | B. Tech | Semester | : | FALL 2025-26 |
| Course Code & Course Title | : | BMAT205L & Discrete Mathematics and Graph Theory | Class Numbers | : | CH2025260100886, CH2025260100887, CH2025260100889, CH2025260100890, CH2025260100892, CH2025260100894, CH2025260100895, CH2025260100896, CH2025260100900, CH2025260100901, CH2025260100902 |
| Faculty | : | Dr. Berin Greeni, Dr. Amit Kumar Rahul, Dr. Soumendu Roy, Dr. Sandip Dalui, Dr. Rajesh Kumar Mohapatra, Dr. Vidhya V, Dr. Hannah Grace G, Dr. Radha S, Dr. Pulak Konar, Dr. Kriti Arya, Dr. Ramesh Babu N | Slot | : | C2+TC2+TCC2 |
| Duration | : | 90 minutes | Max. Mark | | 50 |

**General Instructions:**

- Write only your registration number on the question paper in the box provided and do not write other information
- Only non-programmable calculator without storage is permitted

**Answer all questions**

| Q. No | Sub Sec. | Description | Marks | CO | BT Level |
|---|---|---|---|---|---|
| 1. | (a) | 25 boxes of apples are delivered to a supermarket. The apples are of three different varieties. All the apples in each box are of same variety. Show that among these boxes there are at least 9 containing the same variety. | 3 | 3 | K4 |
| | (b) | Using the generating function, solve for $f_n$ where $f_{n+2} = f_{n+1} + 2f_n$, $n \geq 0$, $f_0 = 2, f_1 = 1$ and hence find $f_2$. | 7 | 3 | K3 |
| 2 | | a) Find the SOP and POS of the Boolean expression $f(x,y,z) = (yz + xz')(xy' + z)'$ using algebraic method.<br>b) Simplify the above Boolean expression and draw the circuit diagram corresponding to it before and after simplifying. | 10 | 4 | K3 |
| 3 | (a) | Draw the Hasse diagram of $\{S_{70}, \leq\}$, where $\leq$ denotes the relation "divisibility" and $S_{70}$ is the set of all divisors of 70. Prove that $\{S_{70}, \leq\}$, is a Lattice. Check whether it is complemented. Justify. | 7 | 4 | K4 |
| | (b) | Show that the pentagonal lattice shown here is not distributive.<br>  | 3 | 4 | K4 |

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| 4 | (a) | Consider the following two graphs G1 and G2.<br>Graph **G1:** Vertices: $V = \{A, B, C, D, E, F\}$and Edges: $\{AB, AC, AD, BE, CF, DE, EF\}$.<br>Graph **G2**: Vertices: $V = \{1,2,3,4,5,6\}$ and Edges: $\{12, 13, 14, 25, 36, 45, 56\}$.<br>Determine whether G1and G2 are isomorphic. If yes, provide a possible vertex mapping between the two graphs. If not, justify why they cannot be isomorphic. | 5 | 5 | K4 |
| | (b) | Consider the graph G with vertices $V = \{A, B, C, D\}$and edges $E = \{AB, AC, AD, BC, CD\}$.<br>a)Draw all possible spanning subgraphs of G with exactly 3 edges.<br>b)Determine which of them are connected. | 5 | 5 | K4 |
| 5 | (a) | A city has 5 locations A, B, C, D, E and road exist between A − B, A − C, A − D, A − E, B − C, B − D, C − E, D − E. Represent this as a graph G. Determine whether G is<br>a) Eulerian<br>b) Hamiltonian.<br>Give proper justifications. | 5 | 5 | K2 |
| | (b) | Give an example of a graph G on 5 vertices which is<br>i) Eulerian but not Hamiltonian<br>ii) Neither Hamiltonian nor Eulerian | 5 | 5 | K2 |

**********All the best **********